

Fragmented Data-Driven Predictive Control: Towards smaller datasets ^{*}

Ruslan Shaiakhmetov ^{*} Danilo Pianini ^{*} Valter Venusti ^{**}
Alessandro V. Papadopoulos ^{***}

^{*} *Department of Computer Science and Engineering,
Alma Mater Studiorum—Università di Bologna, Italy, (e-mail:
ruslan.shaiakhmetov@unibo.it, danilo.pianini@unibo.it)*

^{**} *Digital Innovation and Vehicle Electronics Solutions,
Dallara Automobili S.p.A., Italy, (e-mail: v.venusti@dallara.it)*

^{***} *Mälardalen University, Sweden, (e-mail:
alessandro.papadopoulos@mdu.se)*

Abstract: Data-driven control, exemplified by DeePC, enables controller synthesis directly from data but remains limited when dealing with nonlinear or stochastic dynamics under scarce datasets. Existing extensions such as S-DeePC mitigate these issues via problem decomposition, yet rely on surrogate formulations that restrict horizon flexibility. This paper presents F-DeePC, a fragmented DeePC approach that explicitly builds the constraint matrices and partitions the prediction horizon into independently adjustable segments. This direct formulation enhances both flexibility and numerical efficiency, yielding improved tracking performance especially when data is limited or the system features nonlinearities.

Keywords: Data-driven control, Predictive control, Optimal control.

1. INTRODUCTION

Data-driven control has emerged as a compelling paradigm for control system design (Coulson et al., 2019), providing an alternative to model-based approaches such as Model Predictive Control (MPC) when first-principles models are unavailable or unreliable, yet input-output data are accessible.

Existing data-driven methods are broadly categorized into indirect and direct approaches (Krishnan and Pasqualetti, 2021; Dörfler et al., 2023). Indirect approaches identify an explicit model from data and subsequently design a controller using classical tools. This two-stage pipeline exploits mature identification techniques, but remains sensitive to model inaccuracies. Conversely, direct approaches bypass explicit identification and compute control laws directly from data, which can improve performance when system modeling is difficult or uncertain. Key results include stability and performance-guaranteed formulations grounded in Willems’ fundamental lemma (Willems et al., 2005; Rapisarda and Pal, 2023), which form the theoretical foundation of Data-enabled Predictive Control (DeePC) (Coulson et al., 2019). DeePC has since been applied to domains such as power systems (Huang et al.,

2022; Markovsky et al., 2023) and aerial robotics (Elokda et al., 2021; Bouffard et al., 2012).

Despite its success, extending DeePC to stochastic and nonlinear systems remains challenging. Recent research has addressed these limitations via relaxation (Abolpour et al., 2023; Seuret et al., 2023), regularization (Dörfler et al., 2022), and data-efficiency enhancements (Dörfler et al., 2023). Other approaches optimize the dataset instead via kernel-based selection or compression (Huang et al., 2024; Pillonetto et al., 2014; Alsalti et al., 2024). Notably, replacing a single long trajectory with multiple shorter ones improves numerical conditioning under noisy data (van Waarde et al., 2020).

Building on this line of work, Segmented Data-enabled Predictive Control (S-DeePC) (O’Dwyer et al., 2023) partitions the optimization problem into smaller segments to reduce tracking error and improve robustness under noise. While segmentation offers flexibility and enables the use of smaller datasets, S-DeePC enforces a uniform horizon across all segments, a simplification that eases implementation but restricts controller adaptability.

Motivation. In many applications, the optimal prediction horizon depends jointly on system controllability (Mishra et al., 2021) and reference characteristics. For instance, discontinuous or rapidly changing reference trajectories may require longer horizons to preserve stability and performance. Allowing horizon lengths to vary across segments can better capture system dynamics, improving control accuracy in data-limited or nonlinear settings.

Contribution. This paper introduces Fragmented Data-enabled Predictive Control (F-DeePC), generalizing over

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S-DeePC by enabling variable prediction horizons across fragments while maintaining a direct problem formulation. This work presents: (i) a direct formulation for horizon-splitting in DeePC; (ii) an efficient method to build constraint matrices with fragment-specific horizons; (iii) an open-source Multiple-Input Multiple-Output (MIMO) implementation of F-DeePC; (iv) an evaluation on noisy Linear Time-Invariant (LTI) and nonlinear systems, demonstrating improved tracking and data efficiency.

2. BACKGROUND

2.1 Data-enabled Predictive Control (DeePC)

DeePC is a data-enabled predictive control scheme leveraging a data-driven internal representation of the plant. At each step, it predicts future outputs and selects inputs that keep the predicted trajectory close to the reference, using linear combinations of previously recorded trajectories that are consistent with the current operating point.

Dataset The dataset consists of N past trajectories of length T containing stacked input $\mathbf{u} \in \mathbb{R}^{m \cdot T}$ and output $\mathbf{y} \in \mathbb{R}^{p \cdot T}$ vectors, with m inputs and p outputs arranged as a single column per trajectory for MIMO systems.

Data-driven predictor Each trajectory is partitioned into an initial (past) part and a future part:

$$\begin{aligned} \mathbf{u}_\triangleleft &\in \mathbb{R}^{m \cdot T_\triangleleft}, & \mathbf{y}_\triangleleft &\in \mathbb{R}^{p \cdot T_\triangleleft}, \\ \mathbf{u}_\triangleright &\in \mathbb{R}^{m \cdot T_\triangleright}, & \mathbf{y}_\triangleright &\in \mathbb{R}^{p \cdot T_\triangleright}. \end{aligned} \quad (1)$$

Here, $T_\triangleleft$ matches the recent history leading to the current state, while $T_\triangleright$ is the *prediction horizon*. Following Markovsky and Rapisarda (2008), the predictor enforces:

$$\underbrace{\begin{pmatrix} \mathbf{u}_\triangleleft^1 & \mathbf{u}_\triangleleft^2 & \dots & \mathbf{u}_\triangleleft^N \\ \mathbf{y}_\triangleleft^1 & \mathbf{y}_\triangleleft^2 & \dots & \mathbf{y}_\triangleleft^N \\ \mathbf{u}_\triangleright^1 & \mathbf{u}_\triangleright^2 & \dots & \mathbf{u}_\triangleright^N \\ \mathbf{y}_\triangleright^1 & \mathbf{y}_\triangleright^2 & \dots & \mathbf{y}_\triangleright^N \end{pmatrix}}_{\mathbf{H} \in \mathbb{R}^{T(m+p) \times N}} \underbrace{\begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_N \end{pmatrix}}_{\mathbf{g} \in \mathbb{R}^N} = \underbrace{\begin{pmatrix} \mathbf{u}_\triangleleft^{ini} \\ \mathbf{y}_\triangleleft^{ini} \\ \mathbf{u}_\triangleright \\ \mathbf{y}_\triangleright \end{pmatrix}}_{\mathbf{L} \in \mathbb{R}^{T(m+p)}}. \quad (2)$$

Here, $\mathbf{H}$ stacks N trajectories, $\mathbf{g}$ are their combination weights, and $\mathbf{L}$ concatenates the online initial conditions and the decision variables:

$$\mathbf{u}_\triangleleft^{ini} \in \mathbb{R}^{m \cdot T_\triangleleft}, \quad \mathbf{y}_\triangleleft^{ini} \in \mathbb{R}^{p \cdot T_\triangleleft}. \quad (3)$$

By Willems' Fundamental Lemma (Markovsky et al., 2005; Markovsky and Rapisarda, 2008), any trajectory of the unknown LTI system that is consistent with the data lies in the span of the columns of $\mathbf{H}$. Thus, any feasible $(\mathbf{g}, \mathbf{u}_\triangleright)$ yields a predicted $\mathbf{y}_\triangleright$ via (2).

A length- L time series is converted into $N = L - T$ sub-trajectories using a sliding window of width T ; stacking them produces the Hankel matrix

$$\mathbf{H} = \begin{bmatrix} h_1 & h_2 & \dots & h_{L-T} \\ h_2 & h_3 & \dots & h_{L-T+1} \\ \vdots & \vdots & \ddots & \vdots \\ h_T & h_{T+1} & \dots & h_L \end{bmatrix}, \quad (4)$$

whose columns collect input-output vectors with the initial/future split as in (2).

Control loop Given the predictor, DeePC computes optimal future inputs $\mathbf{u}_\triangleright$ by solving:

$$\begin{aligned} &\underset{\mathbf{g}, \mathbf{u}_\triangleright, \mathbf{y}_\triangleright}{\text{minimize}} && V(\mathbf{g}, \mathbf{u}_\triangleright, \mathbf{y}_\triangleright) \\ &\text{subject to} && \mathbf{H}\mathbf{g} = \mathbf{L}, \quad \mathbf{u}_\triangleright \in \mathcal{U}, \quad \mathbf{y}_\triangleright \in \mathcal{Y}, \end{aligned} \quad (5)$$

where $\mathcal{U}$ and $\mathcal{Y}$ encode input/output constraints. A standard quadratic objective is:

$$V(\mathbf{g}, \mathbf{u}_\triangleright, \mathbf{y}_\triangleright) = \|\mathbf{y}_\triangleright - \mathbf{r}\|_Q^2 + \|\mathbf{u}_\triangleright\|_R^2, \quad (6)$$

with $Q > 0$ and $R > 0$ weighting tracking and effort. The reference $\mathbf{r}$ spans $T_\triangleright$ steps and is updated receding-horizon-wise. The first input in $\mathbf{u}_\triangleright$ is applied, the measurements updated $(\mathbf{u}_\triangleleft^{ini}, \mathbf{y}_\triangleleft^{ini})$, and the procedure repeated.

Linear relaxation and regularization Equation (2) captures noiseless linear behavior (Markovsky et al., 2005; Markovsky and Rapisarda, 2008). To handle noise and modest nonlinearities, DeePC employs a linear relaxation (Coulson et al., 2019) by introducing slack variables $\boldsymbol{\varepsilon}_\triangleleft \in \mathbb{R}^{p \cdot T_\triangleleft}$ and $\boldsymbol{\varepsilon}_\triangleright \in \mathbb{R}^{p \cdot T_\triangleright}$ that bound deviations in initial and predicted outputs. To mitigate overfitting (Coulson et al., 2019), an ℓ_1 regularizer with coefficient λ_g penalizes model complexity (Shafieezadeh-Abadeh et al., 2019), with $\lambda_\triangleleft$ and $\lambda_\triangleright$ commonly set to 1 (O'Dwyer et al., 2016). The problem becomes:

$$\begin{aligned} &\underset{\mathbf{g}, \mathbf{u}_\triangleright, \mathbf{y}_\triangleright, \boldsymbol{\varepsilon}_\triangleleft, \boldsymbol{\varepsilon}_\triangleright}{\text{minimize}} && V(\mathbf{g}, \mathbf{u}_\triangleright, \mathbf{y}_\triangleright, \boldsymbol{\varepsilon}_\triangleleft, \boldsymbol{\varepsilon}_\triangleright) + \\ &&& \lambda_g \|\mathbf{g}\|_1 + \lambda_\triangleleft \|\boldsymbol{\varepsilon}_\triangleleft\|_1 + \lambda_\triangleright \|\boldsymbol{\varepsilon}_\triangleright\|_1 \\ &\text{subject to} && \mathbf{u}_\triangleright \in \mathcal{U}, \quad \mathbf{y}_\triangleright \in \mathcal{Y}, \\ &&& \boldsymbol{\varepsilon}_\triangleleft \geq \mathbf{0}, \quad \boldsymbol{\varepsilon}_\triangleright \geq \mathbf{0}, \\ &&& \begin{bmatrix} \mathbf{0} \\ -\boldsymbol{\varepsilon}_\triangleleft \\ \mathbf{0} \\ -\boldsymbol{\varepsilon}_\triangleright \end{bmatrix} \leq \mathbf{H}\mathbf{g} - \mathbf{L} \leq \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\varepsilon}_\triangleleft \\ \mathbf{0} \\ \boldsymbol{\varepsilon}_\triangleright \end{bmatrix}. \end{aligned} \quad (7)$$

2.2 Segmented DeePC

For long horizons, DeePC may be sensitive to unmeasured disturbances (Alpago et al., 2020; Berberich et al., 2021). S-DeePC addresses this by partitioning the problem into F equal-length segments, each with horizon $T_\triangleleft$. The same segmentation can be applied to the dataset so that each segment serves as a separate trajectory. Consequently, each subproblem effectively accesses $N[T/T_\triangleleft]$ segments rather than N whole trajectories, improving numerical conditioning and prediction accuracy.

Unlike the direct relation (2), S-DeePC uses an *indirect* least-squares predictor P^* (Dörfler et al., 2023):

$$\begin{aligned} P^* &= \arg \min_P \left\| P \begin{bmatrix} \mathbf{u}_\triangleleft^1 & \mathbf{u}_\triangleleft^2 & \dots & \mathbf{u}_\triangleleft^N \\ \mathbf{y}_\triangleleft^1 & \mathbf{y}_\triangleleft^2 & \dots & \mathbf{y}_\triangleleft^N \\ \mathbf{u}_\triangleright^1 & \mathbf{u}_\triangleright^2 & \dots & \mathbf{u}_\triangleright^N \end{bmatrix} - \begin{bmatrix} \mathbf{y}_\triangleright^1 \\ \mathbf{y}_\triangleright^2 \\ \vdots \\ \mathbf{y}_\triangleright^N \end{bmatrix}^\top \right\|_F^2 \\ &= \begin{bmatrix} \mathbf{y}_\triangleright^1 \\ \mathbf{y}_\triangleright^2 \\ \vdots \\ \mathbf{y}_\triangleright^N \end{bmatrix}^\top \begin{bmatrix} \mathbf{u}_\triangleleft^1 & \mathbf{u}_\triangleleft^2 & \dots & \mathbf{u}_\triangleleft^N \\ \mathbf{y}_\triangleleft^1 & \mathbf{y}_\triangleleft^2 & \dots & \mathbf{y}_\triangleleft^N \\ \mathbf{u}_\triangleright^1 & \mathbf{u}_\triangleright^2 & \dots & \mathbf{u}_\triangleright^N \end{bmatrix}^\dagger. \end{aligned} \quad (8)$$

Partitioning P^* as $P_{\mathbf{u}_\triangleleft}^*$, $P_{\mathbf{y}_\triangleleft}^*$, and $P_{\mathbf{u}_\triangleright}^*$ yields

$$\mathbf{y}_\triangleright = P^* \begin{bmatrix} \mathbf{u}_\triangleleft^{ini} \\ \mathbf{y}_\triangleleft^{ini} \\ \mathbf{u}_\triangleright \end{bmatrix} = \begin{bmatrix} P_{\mathbf{u}_\triangleleft}^* & P_{\mathbf{y}_\triangleleft}^* & P_{\mathbf{u}_\triangleright}^* \end{bmatrix} \begin{bmatrix} \mathbf{u}_\triangleleft^{ini} \\ \mathbf{y}_\triangleleft^{ini} \\ \mathbf{u}_\triangleright \end{bmatrix}. \quad (9)$$

Introducing F subproblems of length $T_\diamond$, we replicate the predictor across segments using the Kronecker product:

$$\begin{bmatrix} \mathbf{y}^{f1} \\ \mathbf{y}^{f2} \\ \dots \\ \mathbf{y}^{fF} \end{bmatrix} = I_F \otimes P_{\mathbf{u}_\diamond}^* \begin{bmatrix} \mathbf{u}^{f0} \\ \mathbf{u}^{f1} \\ \dots \\ \mathbf{u}^{fF-1} \end{bmatrix} + I_F \otimes P_{\mathbf{y}_\diamond}^* \begin{bmatrix} \mathbf{y}^{f0} \\ \mathbf{y}^{f1} \\ \dots \\ \mathbf{y}^{fF-1} \end{bmatrix} + I_F \otimes P_{\mathbf{u}_\bullet}^* \begin{bmatrix} \mathbf{u}^{f1} \\ \mathbf{u}^{f2} \\ \dots \\ \mathbf{u}^{fF} \end{bmatrix}. \quad (10)$$

Define the segmented predictor

$$\Phi = [I_F \otimes P_{\mathbf{u}_\diamond}^* \quad I_F \otimes P_{\mathbf{y}_\diamond}^* \quad I_F \otimes P_{\mathbf{u}_\bullet}^*] \\ = [\Phi_{\mathbf{u}_\diamond} \quad \Phi_{\mathbf{y}_\diamond} \quad \Phi_{\mathbf{u}_\bullet}], \quad (11)$$

so that

$$\mathbf{y}_\bullet = \Phi \begin{bmatrix} \mathbf{u}_\diamond^{ini} \\ \mathbf{y}_\diamond^{ini} \\ \mathbf{u}_\bullet \end{bmatrix} = [\Phi_{\mathbf{u}_\diamond} \Phi_{\mathbf{y}_\diamond} \Phi_{\mathbf{u}_\bullet}] \begin{bmatrix} \mathbf{u}_\diamond^{ini} \\ \mathbf{y}_\diamond^{ini} \\ \mathbf{u}_\bullet \end{bmatrix}. \quad (12)$$

With segmentation, the weighting vector expands to $\mathbf{g}^\top = [\mathbf{g}_1^\top \quad \mathbf{g}_2^\top \quad \dots \quad \mathbf{g}_F^\top] \in \mathbb{R}^{NF}$. The S-DeePC optimization problem is

$$\begin{aligned} & \underset{\mathbf{g}, \mathbf{u}_\bullet, \mathbf{y}_\bullet, \boldsymbol{\varepsilon}_\bullet, \boldsymbol{\varepsilon}_\diamond}{\text{minimize}} && V(\mathbf{g}, \mathbf{u}_\bullet, \mathbf{y}_\bullet, \boldsymbol{\varepsilon}_\bullet, \boldsymbol{\varepsilon}_\diamond) + \|\boldsymbol{\varepsilon}_\diamond\|_1 + \|\boldsymbol{\varepsilon}_\bullet\|_1 + \lambda_g \|\mathbf{g}\|_1 \\ & \text{subject to} && \mathbf{u}_\bullet \in \mathcal{U}, \mathbf{y}_\bullet \in \mathcal{Y} \\ & && \boldsymbol{\varepsilon}_\diamond \geq \mathbf{0}, \boldsymbol{\varepsilon}_\bullet \geq \mathbf{0} \\ & && P_{\mathbf{u}_\diamond}^* \mathbf{g}_1 = \mathbf{u}_\diamond^{ini} \\ & && -\boldsymbol{\varepsilon}_\bullet \leq P_{\mathbf{y}_\diamond}^* \mathbf{g}_1 - \mathbf{y}_\diamond^{ini} \leq \boldsymbol{\varepsilon}_\bullet \\ & && P_{\mathbf{u}_\diamond}^* \mathbf{g}_i = P_{\mathbf{u}_\bullet}^* \mathbf{g}_{i-1} \quad \forall i \in [2, F] \\ & && P_{\mathbf{y}_\diamond}^* \mathbf{g}_i = P_{\mathbf{y}_\bullet}^* \mathbf{g}_{i-1} \quad \forall i \in [2, F], \end{aligned} \quad (13)$$

where the last two constraints enforce inter-segment consistency.

3. PROPOSED APPROACH: FRAGMENTED DEEPC

We generalize S-DeePC by removing the requirement that each subproblem use a prediction horizon equal to the length of the initial window ($T_\diamond$). Instead, we allow *fragment-specific* horizons: each subproblem (hereafter, a *fragment*) has its own initial and prediction lengths. This added flexibility improves adaptability to reference content and local dynamics while preserving the data-driven nature of the formulation.

We realize fragmentation by *directly* constructing the global constraint matrix of a quadratic program (Quadratic Programming (QP)) (Stellato et al., 2020):

$$\begin{aligned} & \underset{x}{\text{minimize}} && \frac{1}{2} x^\top Q x + q^\top x \\ & \text{subject to} && b_l \leq A x \leq b_u, \end{aligned} \quad (14)$$

where the stacked decision vector is

$$x^\top = [\mathbf{g}_1^\top \quad \mathbf{g}_2^\top \quad \dots \quad \mathbf{g}_F^\top \quad \mathbf{u}_\bullet^\top \quad \mathbf{y}_\bullet^\top \quad \boldsymbol{\varepsilon}_\diamond^\top \quad \boldsymbol{\varepsilon}_\bullet^\top \quad \boldsymbol{\varepsilon}^{\text{link}\top}], \quad (15)$$

and A aggregates all equality/inequality, relaxation, and inter-fragment connectivity constraints.

3.1 Constructing the FDeePC matrices

Fragmentation model. Let F denote the number of fragments. In the *uniform* case, each fragment has total length

Table 1. Explanation of submatrices in (16).

Submatrix	Description
A_0	Bounds on predicted inputs and slack variables
$A_1 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times FN}$	Dataset H duplicated across fragments: $I_F \otimes H$
$A_2 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times mT_\bullet}$	Inter-fragment input connectivity
$A_3 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times pT_\bullet}$	Inter-fragment output connectivity
$A_4 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times pT_\diamond}$	Initial-condition upper relaxation (slacks)
$A_5 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times pT_\bullet}$	Prediction-horizon upper relaxation (slacks)
$A_6 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times pT_\diamond}$	Initial-condition lower relaxation (slacks)
$A_7 \in \mathbb{R}^{F(m+p)T_\mathcal{F} \times pT_\bullet}$	Prediction-horizon lower relaxation (slacks)

$T_\mathcal{F} := T_\diamond + T_{\mathcal{F}\bullet}$, and step $s := T_{\mathcal{F}\bullet}$. In the *heterogeneous* case, fragment i has length $T_{\mathcal{F},i} := T_\diamond + T_{\mathcal{F}\bullet,i}$ with step $s_i := T_{\mathcal{F}\bullet,i}$, and $\sum_{i=1}^F T_{\mathcal{F}\bullet,i} = T_\bullet$. All selection and connectivity matrices are dimensioned per fragment.

Block structure of A . The global constraint matrix A is assembled by concatenating block rows/columns, each enforcing a specific constraint type:

$$A = \begin{bmatrix} & A_0 & & & & \\ & & & & & \\ & A_1 & A_2 & A_3 & A_4 & A_5 \\ & & & & & \\ & A_1 & A_2 & A_3 & A_6 & A_7 \end{bmatrix}. \quad (16)$$

Each block has the role summarized in Table 1. In the uniform case, $A_0 \in \mathbb{R}^{(p+m)T_\bullet \times (FN + (m+2p)T_\bullet + pT_\diamond)}$.

Dynamic data and connectivity constraints. Block A_1 replicates the Hankel dataset H for all fragments, providing the data-driven dynamics. Blocks A_2 – A_3 enforce *inter-fragment continuity*: the terminal window of fragment i becomes the initial window of fragment $i+1$. We encode this using fixed binary selection/shift matrices M_i acting on the stacked prediction $(\mathbf{u}_\bullet, \mathbf{y}_\bullet)$. For an integer offset i , M_i extracts the $T_\diamond$ -long past window that ends i samples after the start of the prediction stack; negative offsets select the measured past. In the heterogeneous case, offsets are given by cumulative sums of $\{T_{\mathcal{F}\bullet,i}\}$.

Blocks A_4 – A_7 implement the linear relaxation with non-negative slacks for initial and future outputs, with sign structure distinguishing upper vs. lower bounds. Together, these blocks allow overlapping, uniform, or heterogeneous fragmentations with independently chosen horizons.

3.2 Right-hand sides and box bounds

The vectors b_l and b_u codify initial conditions, inter-fragment continuity, and global boxes. We proceed as follows.

(i) *Initial-condition enforcement.* For the portion of the horizon influenced by the measured past, we embed the online initial inputs/outputs $(\mathbf{u}_\diamond^{ini}, \mathbf{y}_\diamond^{ini})$ directly in the corresponding entries of (b_l, b_u) associated with the first fragment's initial window.

(ii) *Continuity across fragments.* Beyond the measured past, we *extend* the initial data over the full prediction horizon by (a) zero-padding inputs and (b) appending the reference trajectory to outputs. A sliding window of

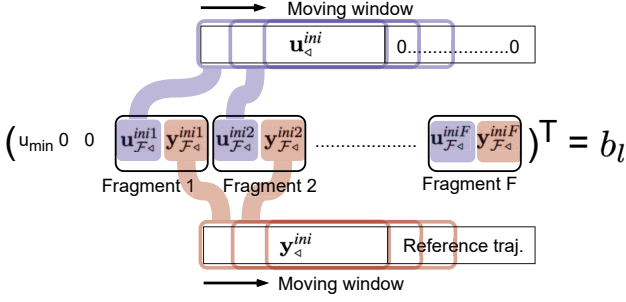


Fig. 1. Constraint-vector construction via a moving window over extended initial data. Zeros in displaced inputs and reference-padded outputs are compensated by the preceding fragment's prediction through the connectivity constraints.

width $T_{\mathcal{F}}$ (or $T_{\mathcal{F},i}$) selects, for each fragment, the relevant slice of these extended vectors. Early fragments thus use true measured data; later fragments use the extended data. Connectivity constraints via M_i then link fragment i to fragment $i+1$ without redundantly re-imposing initial conditions.

(iii) *Global constraints.* We append input bounds $[\mathbf{u}_{\min}, \mathbf{u}_{\max}]$ and any output boxes via the relaxation blocks, yielding the final (b_l, b_u) consistent with (14)–(15).

4. FEASIBILITY AND PRACTICAL STABILITY

This section establishes the theoretical properties of the fragmented DeePC (FDeePC) controller proposed in Section 3. Specifically, we prove (i) feasibility of the QP (14), (ii) its recursive feasibility under terminal ingredients, and (iii) input-to-state *practical* stability (ISpS) of the closed loop.

4.1 Preliminaries and notation

Let $\mathcal{P}_t$ denote the QP (14) at time t , whose decision vector x is stacked as in (15) and whose constraint matrix A is structured as in (16). The feasible set of $\mathcal{P}_t$ is denoted by $\mathcal{F}_t$, and its optimal value by V_t . The cost matrices satisfy $Q \geq 0$, $R > 0$, and, when a terminal penalty is used, $P > 0$. Slack variables represent bounded data mismatch and constraint relaxation.

Assumption 1. (Data richness & bounded mismatch). The identification dataset satisfies persistency of excitation of order T , so that Willems' fundamental lemma applies to all trajectories of length T . The additive prediction mismatch in the FDeePC constraints is bounded and captured by the nonnegative slack variables included in (15).

Assumption 2. (Convex constraints and weights). The admissible input set $\mathcal{U} = \{u : \underline{u} \leq u \leq \bar{u}\}$ is convex and compact. Output constraints are either hard boxes $\underline{y} \leq y \leq \bar{y}$ or soft constraints penalized by nonnegative slacks. The quadratic weights satisfy $R > 0$ and $Q \geq 0$.

Assumption 3. (Terminal ingredients). There exists a compact robust positively invariant (RPI) set $\mathcal{Z}_f \subseteq \{y : \underline{y} \leq y \leq \bar{y}\}$ and a control policy $\kappa : \mathbb{R}^p \rightarrow \mathcal{U}$ such that (a) $\mathcal{Z}_f$ is RPI for the nominal prediction model when slacks vanish, and (b) the terminal cost $V_f(y) = \|y\|_P^2$ is a control Lyapunov function for κ on $\mathcal{Z}_f$.

4.2 Behavioral consistency of fragmentation

Lemma 1. (Behavioral equivalence in the noiseless case). Under Assumption 1, if all slacks in (15) are zero, any $x \in \mathcal{F}_t$ satisfying the connectivity constraints encoded in A generates a single length- T trajectory consistent with the system behavior represented by the dataset Hankel matrix used in A_1 . Equivalently, the concatenation of all fragment trajectories belongs to the behavioral subspace spanned by the original DeePC formulation.

Proof. Each fragment enforces data consistency through its block of A_1 , which reproduces a window of Hankel columns. The inter-fragment connectivity matrices M_i appearing in blocks A_2 – A_3 enforce equality of the overlapping windows. By induction over fragments, the concatenation of all fragment trajectories forms a single admissible trajectory of length T , which lies in the behavior implied by the persistently exciting dataset.

4.3 Feasibility

Theorem 1. (Initial feasibility). Under Assumptions 1 and 2, the QP (14) is feasible for any initial measured window whenever the output constraints are softened by nonnegative slacks.

Proof. Select any constant input sequence $\bar{u} \in \mathcal{U}$ and set $\mathbf{u}_* = \bar{u}$. Choose arbitrary outputs $\mathbf{y}_*$ (since softened). Setting $\mathbf{g}_i = 0$ for all fragments and defining the slacks to satisfy the affine equalities represented in the corresponding block rows of A yields a feasible point. If any slack component is negative, it can be increased without violating bounds, ensuring $x \in \mathcal{F}_t$.

4.4 Recursive feasibility

Theorem 2. (Recursive feasibility). Suppose Assumptions 1–3 hold, and augment (14) with the terminal constraint $y_{T_*} \in \mathcal{Z}_f$. If $\mathcal{F}_t \neq \emptyset$ at time t , then $\mathcal{F}_{t+1} \neq \emptyset$ at time $t+1$.

Proof. Let x_t^* be an optimizer of (14). Construct a candidate for time $t+1$ by shifting the optimal predicted sequences $\mathbf{u}_*^*$ and $\mathbf{y}_*^*$ one step forward and appending $(\tilde{u}, \tilde{y})$, where $\tilde{u} = \kappa(y_{T_*}^*)$ and $\tilde{y} \in \mathcal{Z}_f$ by the RPI property. The new past window corresponds to the last T_d samples of the previous optimal trajectory. By Lemma 1, there exist fragment weights $\tilde{\mathbf{g}}_i$ satisfying the data-consistency equalities in A_1 and the inter-fragment constraints in A_2 – A_3 . Bounded mismatch is absorbed by nonnegative slacks (Assumption 1). Hence the constructed point satisfies all constraints of $\mathcal{P}_{t+1}$, implying $\mathcal{F}_{t+1} \neq \emptyset$.

Remark 1. (Terminal-free case). When terminal ingredients are omitted, recursive feasibility with *hard* output constraints is not guaranteed. With *soft* outputs, the shifted candidate above remains feasible at all times because the nonnegative slack variables can absorb constraint violations in (15).

4.5 Practical stability

We adopt the standard input-to-state practical stability (ISpS) definition: the closed-loop signal $z_t := y_t - r_t$ is

ISpS if there exist $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$, $\sigma \in \mathcal{K}$, and $c \geq 0$ such that:

$$\begin{aligned} \alpha_1(\|z_t\|) &\leq V_t \leq \alpha_2(\|z_t\| + \|u_t\|), \\ V_{t+1} - V_t &\leq -\sigma(\|z_t\|) + c. \end{aligned} \quad (17)$$

Theorem 3. (Input-to-state practical stability). Under Assumptions 1–3, and with $R > 0$, $Q \geq 0$, and $P > 0$, the closed loop induced by sequentially solving (14) satisfies the ISpS inequalities (17). Consequently, the closed-loop trajectories remain bounded, and the bound shrinks with the mismatch and noise levels represented by the slack variables.

Proof. Consider the feasible shifted candidate used in the proof of Theorem 2. Because $R > 0$ and $P > 0$, the stage cost and terminal cost are positive definite in (u, y) , implying a coercive value function V_t . Evaluating $J(x_{t+1})$ for the shifted candidate shows that $V_{t+1} - V_t$ decreases by at least the stage cost difference, up to a bounded term proportional to the slack norms (representing mismatch). Bounding these terms by suitable class- $\mathcal{K}$ functions yields the ISpS inequalities, as in standard terminal-set MPC arguments.

Corollary 1. (Terminal-free practical stability). If the terminal ingredients in Assumption 3 are omitted but the total prediction horizon in (14) exceeds a problem-dependent threshold T^* , then, under soft output constraints, the same ISpS inequalities hold with constants depending on T_* . The residual bound c tends to zero as the mismatch magnitude and slack usage vanish.

Proof. With softened outputs, recursive feasibility follows for all t . For sufficiently long horizons, the standard value-function decrease argument for horizon-based MPC implies (17) up to a bounded remainder inversely proportional to T_* , which is absorbed in c .

5. EXPERIMENTAL EVALUATION

The proposed approach and state-of-the-art methods have been experimentally evaluated on two Single-Input Single-Output (SISO) systems: a LTI system (a two-mass-spring-damper) with noise, and a nonlinear system (double pendulum). The latter, in particular, significantly challenges the control algorithms under evaluation due to its nonlinear and chaotic behavior.

Noisy LTI system. The LTI system, depicted in Figure 2, consists of two masses $m_1 = 0.5\text{kg}$ and $m_2 = 1.5\text{kg}$ connected by two springs $k_1 = k_2 = 2\text{N/s}$ and dampers $c_1 = 1\text{kg/s}$ and $c_2 = 1\text{kg/s}$. This is the same reference system used in (O’Dwyer et al., 2023), which we are reusing to compare performance as fairly as possible. The state-space representation of the system is:

$$\begin{aligned} \dot{x} &= Ax + B(u + n_{\text{rand}} + n_{\text{sin}}) \\ y &= Cx + n_{\text{meas}} \end{aligned} \quad (18)$$

where $x^\top = [x_1, x_2, \dot{x}_1, \dot{x}_2]$ is the vector of positions and velocities, $u \in [-1\text{N}, 1\text{N}]$ is the input force, $n_{\text{rand}} \sim \mathcal{U}(-0.15, 0.15)\text{N}$ is a uniform noise, $n_{\text{sin}} \sim 0.2 \cdot \sin(2\pi \cdot 0.01 \cdot t)\text{N} + 0.2\text{N}$ is a sinusoidal noise, and $n_{\text{meas}} \sim \mathcal{N}(0, 0.1)\text{m}$ is a normally-distributed noise.

Nonlinear system. We study a double pendulum with damping at the interconnecting joint, depicted in Figure 3. The damping coefficient $\zeta \in \mathbb{R}_0^+$ control the

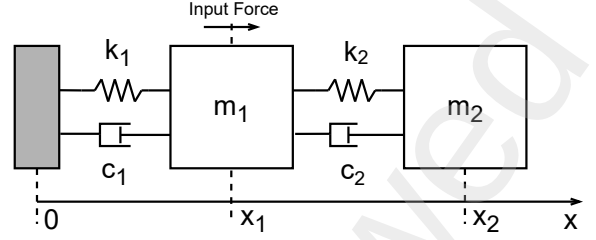


Fig. 2. Linear Time-Invariant system for evaluation of the proposed approach. It is a two-mass-spring-damper system with a single input and single output. The input force is applied to the first mass along the x -axis, and the output is the position of the second mass.

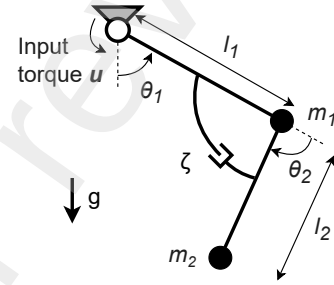


Fig. 3. Mechanical scheme of the double pendulum system with a damped interconnecting joint.

system’s behavior, with $\zeta \rightarrow \infty$, reducing the system to a single pendulum. In our experiments, we use $\zeta \in [0.125, 0.25, 0.5, 1.0]\text{Nms}$, as we found empirically that $\zeta = 1\text{Nms}$ produces a system simple enough for all algorithms under test to control. Figure 4 illustrates how system ζ influences the system dynamics.

Evaluation metric. We measure the performance of all algorithms under test using the Sum of Set-Point Tracking Errors (SPTE), used in (O’Dwyer et al., 2023), which measures the accumulated absolute error between the system position and the reference trajectory.

Reproducibility. We repeat every experiment 100 times to ensure statistical significance. For each repetition, we use a dedicated random generator seeded with the repetition number to ensure reproducibility. The random generator used for each repetition is used for the generation of the shared dataset and, in the noisy LTI system, for the generation of the noise.

The implementation of S-DeePC and its original unsegmented version are from the companion code of (O’Dwyer et al., 2023)¹. The implementation of the proposed approach has been written in Python using NumPy (Harris et al., 2020) and the OSQP quadratic programming solver (Stellato et al., 2020). Our reference implementation of F-DeePC has been released as open-source (MIT-licensed) software² and has been archived for future reference on Zenodo (Shaiakhmetov, 2025).

¹ <https://codeocean.com/capsule/3272665/tree>

² <https://github.com/Crylab/DeepC-Fragmented>

Step Response of Double Pendulum for Varying Damping Coefficients

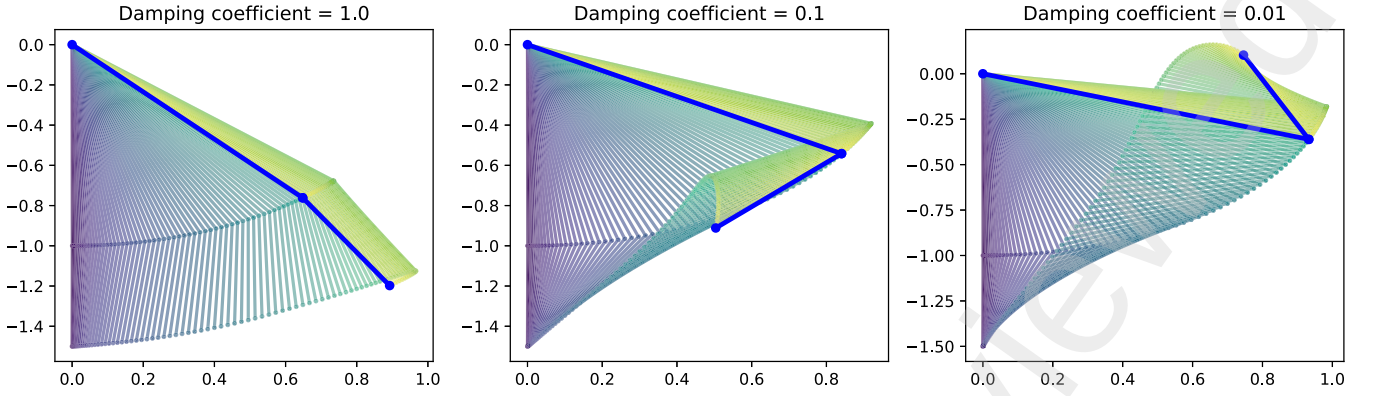


Fig. 4. Step response of a double pendulum to a constant torque on its first joint, with varying damping coefficients: $\zeta = 1\text{N m s}$ (left), 0.1N m s (center), and 0.01N m s (right). Lower damping values lead to more chaotic responses.

5.1 Hyperparameters analysis

All the evaluated algorithms feature a **prediction horizon**, an **initial horizon**, and a **regularization term** λ_g . F-DeePC additionally introduces a **fragmented prediction horizon** $T_{\mathcal{F}}$.

While DeePC and S-DeePC are largely insensitive to λ_g (O'Dwyer et al., 2023), the response of F-DeePC requires further investigation. Similarly, the effect of different values of $T_{\mathcal{F}}$ must be explored.

To analyze the influence of these two hyperparameters in F-DeePC, we evaluate set-point tracking performance on a noiseless LTI system, consistent with the rest of the study. Each configuration is tested over 30 repetitions, with different random realizations of the dataset. The results, shown in Figure 5, indicate that the best tracking performance occurs when $T_{\mathcal{F}}$ is set to 3 steps, which aligns with the system's controllability. Specifically, a fragmented prediction horizon of 3 steps provides a sufficient prediction length, ensuring that the system remains within the reference neighborhood (Mishra et al., 2021). The regularization term λ_g has little impact on performance, consistent with the behavior observed in DeePC and S-DeePC. For the remainder of the evaluation, we set $\lambda_g = 6.5$ and $T_{\mathcal{F}} = 3$.

5.2 LTI system with noise

Dataset Generation. The dataset for all algorithms under test is generated once using a discretized version of the LTI system with a timestep of $\Delta t = 1\text{s}$. The system starts from zero initial conditions and is driven by a persistently excited (Willems et al., 2005) input force in $[-1\text{N}, 1\text{N}]$ whose value changes randomly every 10 time steps. After collecting L samples, the data is stored in a Hankel matrix, as shown in Equation 4.

Performance Evaluation. Results are summarized in Figure 6. The case $L = 50$ is detailed in Figure 7. As expected, performance improves with larger datasets, however, S-DeePC show better error reduction rate than F-DeePC due to its indirect formulations (akin to DeePC), which is known to perform better with high noise-to-signal

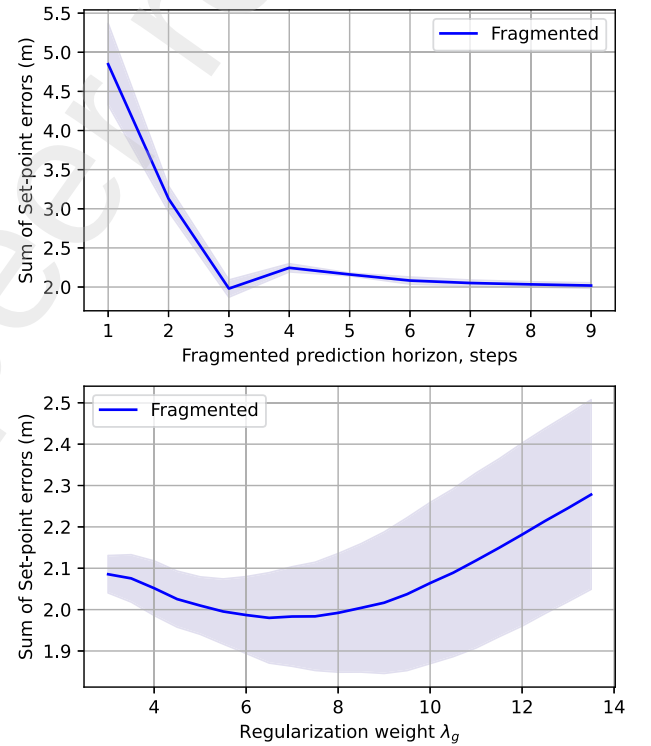


Fig. 5. Fragmented prediction horizon $T_{\mathcal{F}}$ (top) and regularization term λ_g (bottom) evaluation on the noiseless LTI system. Shaded areas represent ± 1 standard deviations. Results are the average of 30 repetitions. $T_{\mathcal{F}}$ has a significant impact, with the best tracking performance at 3 steps. λ_g influences performance weakly, with the optimum value at 6.5.

ratios (Dörfler et al., 2023; Krishnan and Pasqualetti, 2021). Nonetheless, the proposed approach outperforms state-of-the-art methods on smaller datasets: DeePC becomes unstable for $L = 40$ and S-DeePC struggles with $L = 30$, while F-DeePC remains stable due to its finer division of the prediction horizon. For $L = 50$, F-DeePC and S-DeePC perform similarly, and both outperform DeePC.

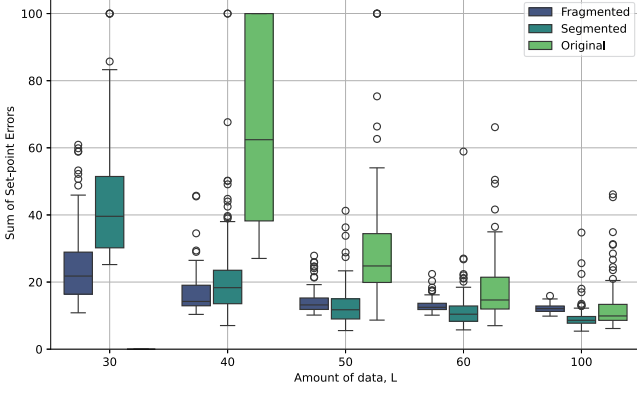


Fig. 6. Boxplots³ of SPTE for an LTI with varying dataset sizes. Fragmented performs best on small datasets, matches Segmented for $L = 50$, and worsens for larger ones.

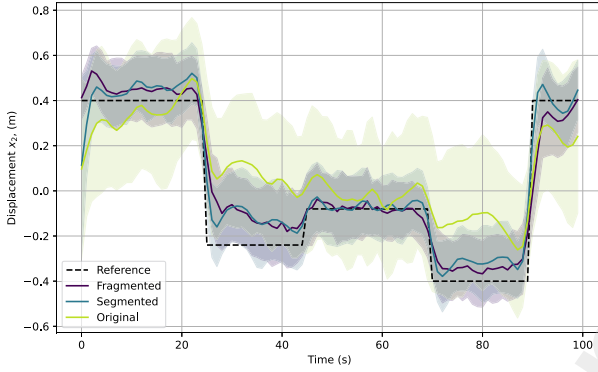


Fig. 7. Tracking performance with $L = 50$. Solid lines show mean trajectories over 100 runs; shaded areas are mean $\pm 1\sigma$, the black dashed line is the reference. Fragmented and Segmented behave similarly, both better than the Original.

5.3 Nonlinear system

Dataset generation. In the case of an LTI system, a persistently exciting input is sufficient to generate a dataset that captures the system dynamics. The same strategy is not viable for nonlinear systems (Willems et al., 2005) in which, however, data can be generated through feedback linearization (Alsalti et al., 2023; Persson et al., 2024). To achieve feedback linearization, we apply the following PD controller with gravity compensation:

$$u_{PD}(\theta_1, \dot{\theta}_1, \theta_2) = l_1 \cdot m_1 \cdot g \cdot \sin(\mathbf{r}) + m_2 \cdot g(l_1 \sin(\mathbf{r}) + l_2 \sin(\theta_2)) + 20(\mathbf{r} - \theta_1) - 10\dot{\theta}_1. \quad (19)$$

We then use random set-points as reference trajectories, each lasting 10 steps, and record the transition and stabilization dynamics. Importantly, the PD auxiliary controller is used only for dataset generation and does not participate in the optimization process. The dataset is generated by simulating the system for L time steps $\Delta t = 0.2s$ and recording the state of the system and the control signal. The dataset is then organized into a Hankel matrix as described in Equation 4.

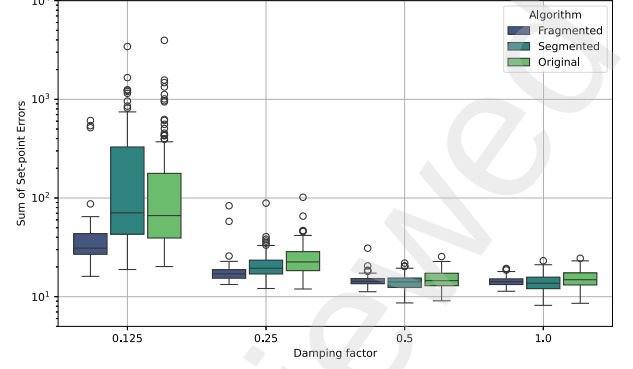


Fig. 8. Boxplots³ of SPTE of a double pendulum with $L = 100$ with different damping ζ . F-DeePC achieves lower median error and variance, particularly in more difficult cases.

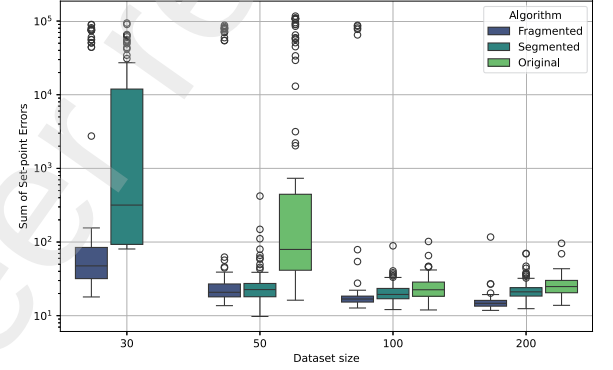


Fig. 9. Boxplots³ of SPTE with $\zeta = 0.25N \text{ ms}$ with and different dataset sizes. F-DeePC outperforms other methods across all dataset sizes.

Performance evaluation. Results for different damping factors are summarized in boxplots³ in Figure 8. Lower damping leads to more chaotic behavior and poorer control performance across all methods, but F-DeePC consistently achieves lower median error and variance, particularly in the most challenging scenarios. A finer prediction horizon splitting improves its ability to capture the system dynamics. As the system becomes simpler, performance differences narrow, although F-DeePC remains superior.

Figure 9 shows the impact of dataset size. Additional data benefits F-DeePC the most, reducing error faster than the baselines. S-DeePC performs worse here for the same reasons performed well in LTI systems: its least squares estimator treats nonlinearities as noise. Overall, F-DeePC achieves the best results across both small and large datasets.

6. CONCLUSION

This work introduced F-DeePC, a novel method that partitions the prediction horizon into smaller fragments based on a direct formulation of the DeePC algorithm. By explicitly constructing the constraint matrix, F-DeePC reduces the dataset requirements and median error while

³ Boxplots show medians, and 1.5 IQR whiskers; outliers are circles

allowing independent adjustment of the horizon lengths in each fragment.

Experiments on a noisy LTI and a damped double pendulum show that F-DeePC consistently outperforms state-of-the-art methods for small datasets. Unlike the indirect formulation of S-DeePC, F-DeePC adopts a direct formulation, trading higher sensitivity to large noise-to-signal ratios in LTIs (Dörfler et al., 2023) with better performance for complex nonlinear systems (Krishnan and Pasqualetti, 2021) across all dataset sizes.

In conclusion, F-DeePC is a promising solution to apply data-driven control in systems where data is limited. Future research will focus on MIMO systems and practical applications in autonomous vehicles and quadrotors.

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